

DATASET INSPECTION REPORT

Dataset: Telco Customer Churn Dataset

<https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Introduction

The dataset inspection was conducted as part of the Week 1 Machine Learning Internship assignment for the Customer Churn Prediction Project. The dataset used is the Telco Customer Churn Dataset, which contains customer information and a target variable indicating whether each customer churned. The assignment requires the dataset to be inspected for data types, missing values, and duplicate records.

Dataset Overview

The dataset contains 7,043 customer records and 21 columns. The columns contain information about customer demographics, services, account information, billing information, and customer churn.

The target variable is "Churn", which indicates whether a customer left the company.

The dataset contains the following columns:

- "customerID"
- "gender"
- "SeniorCitizen"
- "Partner"
- "Dependents"
- "tenure"
- "PhoneService"
- "MultipleLines"
- "InternetService"
- "OnlineSecurity"
- "OnlineBackup"
- "DeviceProtection"
- "TechSupport"
- "StreamingTV"
- "StreamingMovies"
- "Contract"
- "PaperlessBilling"
- "PaymentMethod"

- "MonthlyCharges"
- "TotalCharges"
- "Churn"

Data Type Inspection

The inspection identified three main data types:

Data type	Number of Columns
Object	17
Integer	2
Float	2

The categorical variables are mainly stored as "object" data types, while "SeniorCitizen" and "tenure" are stored as integers. "MonthlyCharges" and "TotalCharges" are stored as floating-point numerical variables.

The "TotalCharges" column was found to have a "float64" data type after conversion to a numerical format.

Missing Value Inspection

Missing-value analysis was performed using the "isnull()" method.

The inspection showed that most columns contain no missing values. However, the "TotalCharges" column contains 11 missing values.

Column	Missing Values
"TotalCharges"	11
All other columns	0

The 11 affected records were examined further. The customers with missing "TotalCharges" have a tenure of 0 months, while their "MonthlyCharges" values are available. Because these records have zero tenure and therefore have no accumulated charges represented in the dataset, the missing "TotalCharges" values will be treated as 0 during preprocessing.

The missing values represent approximately 0.16% of the entire dataset, so their proportion is very small.

Duplicate Inspection

Two forms of duplication were investigated.

- Duplicate Rows

The following check was performed:

```
dataset.duplicated().sum()
```

The result was 0, indicating that there are no completely duplicated records in the dataset.

- Duplicate Customer IDs

The "customerID" column was also checked for repeated customer identifiers.

The result was 0, indicating that each customer ID is unique and no customer appears more than once based on the identifier.

Categorical Variable Inspection

The categorical variables were examined to identify their unique categories and frequency distributions.

- The "gender" variable contains two categories:

- Male: 3,555

- Female: 3,488

- The "Partner" variable contains:

- No: 3,641

- Yes: 3,402

- The "Dependents" variable contains:

- No: 4,933

- Yes: 2,110

- The "PhoneService" variable contains:

- Yes: 6,361

- No: 682

- The "MultipleLines" variable contains three categories:

- No: 3,390

- Yes: 2,971

- No phone service: 682

- The "InternetService" variable contains:

- Fiber optic: 3,096

- DSL: 2,421

- No: 1,526

The service-related variables such as "OnlineSecurity", "OnlineBackup", "DeviceProtection", "TechSupport", "StreamingTV", and "StreamingMovies" contain the category "No internet service". This is a valid category rather than a missing value because it indicates that the customer does not have internet service.

- The "Contract" variable contains:
 - Month-to-month: 3,875
 - Two year: 1,695
 - One year: 1,473
- The "PaymentMethod" variable contains four categories:
 - Electronic check: 2,365
 - Mailed check: 1,612
 - Bank transfer (automatic): 1,544
 - Credit card (automatic): 1,522

Target Variable Inspection

The target variable is "Churn".

Its distribution is:

Churn	Number of Customers	Approx. Percentage
No	5,174	73.5%
Yes	1,869	26.5%
Total	7,043	100%

The target variable is therefore moderately imbalanced, because the number of customers who did not churn is considerably greater than the number who churned.

This imbalance must be considered when selecting model evaluation metrics. Accuracy alone may not provide a complete assessment of model performance, so metrics such as precision, recall, F1-score, and ROC-AUC should also be considered during model evaluation.

Numerical Variable Inspection

Descriptive statistics were calculated for the four numerical variables: "SeniorCitizen", "tenure", "MonthlyCharges", and "TotalCharges".

Variable	Mean	Median	Minimum	Maximum
"SeniorCitizen"	0.1621	0	0	1
"tenure"	32.3711	29	0	72
"MonthlyCharges"	64.7617	70.3500	18.25	118.75
"TotalCharges"	2279.7343	1394.5500	0	8684.80

"SeniorCitizen" is stored as an integer but represents a binary indicator, where 0 represents a non-senior customer and 1 represents a senior customer. Its mean of approximately 0.162 indicates that about 16.2% of customers are senior citizens.

The "tenure" variable ranges from 0 to 72 months. Its mean is approximately 32.37 months, while its median is 29 months, indicating some variation in the length of customer relationships.

"MonthlyCharges" ranges from 18.25 to 118.75, with a mean of approximately 64.76 and a median of 70.35. The difference between the mean and median suggests that the distribution may not be perfectly symmetric and should be examined further using a histogram.

"TotalCharges" ranges from 0 to 8,684.80. Its mean of approximately 2,279.73 is considerably higher than its median of 1,394.55, suggesting that the variable may be right-skewed due to customers with relatively high accumulated charges.

The numerical distributions were examined further during Exploratory Data Analysis using histograms and other visualizations.

Key Findings

The dataset inspection produced the following findings:

1. The dataset contains 7,043 records and 21 columns.
2. There are 17 categorical/object columns, 2 integer columns, and 2 floating-point columns.
3. There are 11 missing values, all located in "TotalCharges".
4. The affected "TotalCharges" records have a tenure of 0 months and valid "MonthlyCharges" values.
5. The 11 missing "TotalCharges" values will be treated as 0 during preprocessing.
6. There are no completely duplicated rows.
7. There are no duplicate customer IDs.
8. Several service-related variables contain "No internet service", which is a valid category rather than missing data.
9. The target variable "Churn" contains 5,174 "No" observations and 1,869 "Yes" observations.
10. The target variable is moderately imbalanced, with approximately 73.5% non-churned customers and 26.5% churned customers.

Correlation Analysis

- A correlation analysis was performed on the numerical variables "SeniorCitizen", "tenure", "MonthlyCharges", and "TotalCharges".
- The strongest correlation was observed between "tenure" and "TotalCharges", with a correlation coefficient of 0.826. This indicates a strong positive relationship, meaning that customers with longer tenure generally tend to have higher accumulated charges.
- "MonthlyCharges" and "TotalCharges" also showed a moderately strong positive correlation of 0.651.

This suggests that customers with higher monthly charges generally tend to have higher total charges.

- The correlation between "tenure" and "MonthlyCharges" was 0.248, indicating a weak positive relationship. Similarly, "SeniorCitizen" had a weak positive correlation with "MonthlyCharges" (0.220) and a very weak positive correlation with "TotalCharges" (0.103).
- The correlation between "SeniorCitizen" and "tenure" was 0.017, indicating almost no linear relationship between the two variables.
- Overall, the strongest relationship in the numerical variables was between "tenure" and "TotalCharges". This relationship should be considered during feature analysis and model development because the two variables contain substantially related information. However, correlation alone is not sufficient reason to remove a feature.

Conclusion

The dataset is generally clean and suitable for further analysis and machine learning preparation. The main data-quality issue identified is the 11 missing values in "TotalCharges". These records were investigated and found to correspond to customers with zero tenure.

There are no duplicate rows or duplicate customer IDs. The categorical variables contain meaningful categories, including "No internet service", which should not be treated as missing data.

The next stage will involve exploratory data analysis and problem framing, including visualization of important variables, identification of relationships between customer characteristics and churn, and preparation of the dataset for machine learning model development.